

MODULE : 2 and 3

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1. What is exploratory testing ?

Answer : Though the current trend in testing is to push for automation, exploratory testing is a new way of thinking. Exploratory testing is a concurrent process where testing design, executive and logging happen simultaneously. The focus of exploratory testing is more on testing as a “thinking” activity.

2.What is the traceability matrix ?

Answer : Test conditions should be able to be linked back to their sources in the test basis, this is known as traceability. To protect against changes you should be able to trace back from every system component to the original requirement that caused its presence.

3. What is boundary value testing ?

Answer: Boundary value analysis is a methodology for designing test cases that concentrates software testing effort on cases near the limit of same validation.

4.What is equivalence Partitioning testing ?

Answer : Aim is to treat groups of inputs as equivalence and to select one representative input to test them all. EP can be used for all levels of testing.

5.What is Integration testing ?

Answer : Integration testing is performed to expose defects in the interfaces and in the interactions between integrated components or systems. Integration Testing is a level of the software testing process where individual units are combined and tested as a group.

6.What determines the level of risk ?

Answer : 1) when testing does find defects , the Quality of the software system increases when those defects are fixed

2) The Quality of the system can be improved through lessons learned from previous projects.

3) analysis of root causes of defects found in other projects can lead to process improvement

4) Process improvement can prevent those defects occurring

5) which in turn, can be improve the quality of future systems

6) testing should be integrated as one of the Quality assurance activities

7.What is Alpha testing ?

Answer : It is always performed by the developers at the software development site.It is always performed in a virtual Environment.It is the form of acceptance testing.

8. What is beta testing ?

Answer : It is always performed by the customers at their own site. It is not performed by an independent testing team. It is performed in a real time environment.

9.what is Component testing ?

Answer: A minimal software item that can be tested in isolation. It means “a unit is the smallest testable part of software” The testing of individual software components.

10.What is functional system testing ?

Answer : A requirement that specifies a function that a system or system component must perform.

11. What is Non functional testing ?

Answer : testing the attributes of a component or system that do not relate to functionality e.g., reliability, efficiency , usability , interoperability , maintainability and portability.

12.What is GUI testing ?

Answer : Graphical user interface testing is the process of testing the system's GUI of the system under test. GUI testing involves checking the screens with the controls like menu , buttons , icons, and all types of bars - toolbar , menu bar , dialog boxes and window etc.

13.What is Adhoc testing ?

Answer: The error guessing is a technique where the experienced and good testers are encouraged to think of situations in which the software may not be able to cope.

14.What is load testing ?

Answer : It's a performance testing to check system behaviour under load. Testing an application under heavy loads, such as testing a website under a range of loads to determine at what point the system's response time degrades or fails.

15.What is stress testing ?

Answer : System is stressed beyond its specifications to check how and when it fails. Performed under heavy load like putting large numbers beyond storage capacity, complex database queries, continuous input to system or database load.

16.What is white box testing and list the types of white box testing ?

Answer : Testing based on an analysis of the internal structure of the components or system. White box testing is also known as “glass testing”
List the type of white box testing:

- 1)statement coverage
 - 2) decision coverage
 - 3) condition coverage
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17.What is black box testing ? What are the different black box testing techniques?

Answer : Testing, either functional or nonfunctional, without reference to the internal structure of the component or system.

There are four specifications-based or black-box technique

- 1)Equivalence Partitioning
 - 2) boundary value analysis
 - 3) decision tables
 - 4) state transition testing
 - 5) Use-case testing
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18. Mentioned what are the categories of defects ?

Answer : there are 5 type of defects

- 1) Data base Quality / Database defects
 - 2) Critical functionality defects
 - 3) Functionality defects
 - 4) Security defects
 - 5) User interface defects
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19. Mention what big bang testing is ?

Answer : In big bang integration testing all components or modules is integrated simultaneously, after which everything which is tested as a whole.

20. What is the purpose of exit criteria ?

Answer :

- 1) Thoroughness - i.e. coverage or requirements
 - 2) Cost or time constraints
 - 3) Percentage of tests run without incident
 - 4) Number of faults remaining
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21. Different between QA vs QC vs Tester

Answer:

QA	QC	TESTER
It is a subset of software test life cycle	QC can be considered as the subset of Quality Assurance.	Testing is the subset of Quality Control.
Preventive activities	It is a corrective process	It is a preventive process.
Process oriented activities	Product oriented activities	Product oriented activities
Focuses on process and procedures rather than conducting actual testing on the system	Focus on actual testing by executing software with intend to identify bug/defect through implementation of procedures and process	Focuses on actual testing.
Activities which ensure the implementation of processes, procedures	Activities which ensure verification of developed software	Activities which ensure the identification of bugs/ error/ defects in

QA	QC	TESTER
It is a subset of software test life cycle	QC can be considered as the subset of Quality Assurance.	Testing is the subset of Quality Control.
and standards in context to verification of development software and intended requirements.	with respect to documented requirements.	the software.

22.When should “Regression Testing” be performed ?

Answer :

- 1) Changed in requirements code is modified according to the requirement
 - 2) New feature is added to the software
 - 3) Defect fixing
 - 4) Performance issue fix
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23.What are the 7 key principles ? Explain in detail ?

Answer :

- 1) **Testing shows presence of defects** : testing reduces the probability of undiscovered defects remaining in the software but, even if no defects are found, it is not a proof of correctness.
- 2) **Exhaustive testing is impossible** : Testing everything including all combinations of inputs and preconditions is not possible.

- 3) **Early testing** : Testing activities should start as early as possible in the development life cycle
 - 4) **Defect Clustering** : A small number of modules contain most of the defects discovered during pre-release testing, or are responsible for the most operational failures.
 - 5) **Pesticide paradox** : If the same tests are repeated over and over again, eventually the same set of test cases will no longer find any new defects.
 - 6) **Testing is context dependent** : different kinds of sites are tested differently
 - 7) **Absence of error fallacy** : even after defects have been resolved it may still be unusable and/or does not fulfill the users needs and expectations
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24.Different between smoke and sanity ?

Answer :

Smoke testing	Sanity testing
Smoke testing is performed to ascertain that the critical functionalities of the program are working fine.	Sanity testing is done to check the new functionality / bugs have been fixed
The objective of this testing is to verify “stability” of the system in order to the with more rigorous testing	The objective of the testing is to verify the “ Rationality” of the system in order proceed with more rigorous testing

Smoke testing is usually documented is scripted	Sanity testing is usually not documented and unscripted
This testing is performed by the developers	Sanity testing is usually performed by testers
Smoke testing is a subset of Regression testing	Sanity testing is subset of acceptance testing
Smoke testing exercise the entire system from end to end particular component	Sanity testing exercise only the of the entire system
Smoke testing is like general health check up	Sanity testing is like specialized health check up

26. Difference between verification and validation ?

Answer :

Verification	Validation
The process of evaluating work-products of a development phase to determine whether they meet the specified requirements for that phase.	The process of evaluating software during of the development process to determine whether it satisfies specification business requirements
To ensure that the product is being built according to the requirements and design specifications , In other words, to ensure that work products meet their specified requirements.	To ensure that the product actually meets the user's needs and that the specifications were correct in the first place. In other words to demonstrate that the product fulfills its intended use When placed in its intended environment.

Are we building the product right ?	Are we building the right product ?
Plans, requirement species, Design, specs, code, test cases	The actual product/ software.
-Reviews -Walkthrough -Inspections	-Testing

27. Explain types of performance testing.

Answer :

1) load testing : It's performance testing to check system behaviour under load. Testing an application under heavy loads, such as testing a website under a range of loads to determine at what point the system's response time degrades or fails.

2) Stress testing : System is stressed beyond its specifications to check how and when it fails. Performed under heavy load like putting large numbers beyond storage capacity, complex database queries, continuous input to system or database load.

3)endurance testing

4) spike testing

5) volume testing

6) scalability testing

28.what is Error , Defect , Bug , and failure?

Answer :

Error : A mistake in coding is called error

Defect : error found by tester is called defect

Bug : defect accepted by the developer team then it is called bug
Failure : build does not meet the requirements then it is called failure

29.what is bug life cycle ?

Answer: “ A computer bug is an error, flaw, mistake, failure or fault in a computer program that prevents it from working correctly or produces an incorrect result. Bugs arise from mistake and errors made by people, in either a program’s source code or its design”

30. Different between functional testing and non functional testing

Answer :

Functional Testing	Non functional Testing
Easy to do manual testing	Tough to do manual testing
Functional testing describe what the product does	Non functional testing describe how good the product works
Business requirements are the inputs to functional testing	Performance parameters like speed, scalability are inputs to non functionality testing
Manual testing or automation tools can be used for functional testing	Using tools will be effective for this testing
Functional testing is executed first	Non functional testing should be performed after functional testing
Functional testing is performed using the non functional functional	Testing checks the performance , reliability , scalability , and other

specification provided by the client and verifies the system against the functional requirements	non functional aspects of a software system.
Types of functional testing are : 1) unit testing 2) smoke testing 3) sanity testing 4) integration testing 5) white box testing 6) black box testing 7) user acceptance testing 8) regression testing	Types of nonfunctional testing are : 1) performance testing 2) load testing 3) volume testing 4) stress testing 5) security testing 6) installation testing 7) penetration testing 8) Compatibility testing

31. Different between SDLC and STLC

<u>SDLC</u>	<u>STLC</u>
Software development life cycle	Software tester life cycle
SDLC is structure imposed on the development of a software product	STLC is structure imposed on the development software testing
Phase of SDLC 1) Requirements collection 2) Analysis 3) Design 4) Implementation 5) Testing 6) Maintenance	Phase of STLC 1) Requirements analysis 2) Test planning 3) Test case development 4) Test environment setup 5) Test execution 6) Test cycle closer

32. What is the difference between test scenario, test cases, test script ?

Answer : Scenario means any functionality that can be tested. It is also called test condition, or test possibility.

Test cases involve the set of steps, conditions and inputs which can be used while performing the testing tasks.

Test script is a set of instructions that will be performed on the system under test to test the system functions as expected.

33.explain what the test plan is ?What is the information that should be covered?

Answer : A document describing the scope, approach, resources and schedule of intended test activities.

All projects require a set of plans and strategies which define how the testing will be conducted.

- 1) Test policy
 - 2) Master test plan / test strategy
 - Functional test plan
 - System integration test plan
 - UAT test plan
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34. Write a scenario of only whatsapp chat messages

Answer :

- 1) Verify that the user can set a chat wallpaper
- 2) Verify that the user can disable/delete his Whatsapp account.

- 3) Verify that the user sets privacy settings like turning on/off last seen, online status, read receipts, etc.
 - 4) Verify that the user can take the complete chat backup of his chats.
 - 5) Verify that the user can check data usage by images, audio, video, and documents in WhatsApp chats.
 - 6) Verify that the user can update notification settings
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35. Write a scenario of Pen

Answer :

- Verify that the length and the diameter of the pen are as per the specifications.
- Verify the outer body material of the pen. Check if it is metallic, plastic, or any other material specified in the requirement specifications.
- Check the color of the outer body of the pen. It should be as per the specifications.
- Verify that the brand name and/or logo of the company creating the pen should be clearly visible.
- Verify that any information displayed on the pen should be legible and clearly visible.
- Verify the type of pen, whether it is a ballpoint pen, ink pen, or gel pen.
- Verify that the user is able to write clearly over different types of papers.
- Check the weight of the pen. It should be as per the specifications. In case not mentioned in the specifications, the weight should not be too heavy to impact its smooth operation.
- Verify if the pen is with a cap or without a cap.
- Verify the color of the ink on the pen.
- Check the odor of the pen's ink on writing over a surface.
- Verify the surfaces over which the pen is able to write smoothly apart from paper e.g. cardboard, rubber surface, etc.
- Verify that the text written by the pen should have consistent ink flow without leaving any blob.
- Check that the pen's ink should not leak in case it is tilted upside down.
- Verify if the pen's ink should not leak at higher altitudes.
- Check how fast the user can write with the pen over supported surfaces.

- Verify the performance or the functioning of a pen when used continuously without stopping
 - Verify the number of characters a user can write with a single refill in the case of ballpoint & gel pens and with full ink, in the case of ink or fountain pens.
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36.write a scenario of Door

Answer :

- Verify if the door is a single door or bi-folded door.
 - Check if the door opens inwards or outwards.
 - Verify that the dimension of the doors are as per the specifications.
 - Verify that the material used in the door body and its parts is as per the specifications.
 - Verify that the color of the door is as specified.
 - Verify if the door is a sliding door or rotating door.
 - Check the position, quality and strength of hinges.
 - Check the type of locks in the door.
 - Check the number of locks in the door interior side or exterior side.
 - Verify if the door is having a peek-hole or not.
 - Verify if the door has a stopper or not.
 - Verify if the door closes automatically or not – spring mechanism.
 - Verify if the door makes noise when opened or closed.
 - Check the door condition when used extensively with water.
 - Check the door condition in different climatic conditions- temperature, humidity etc.
 - Check the amount of force- pull or push required to open or close the door.
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37. Write a scenario of an ATM.

Answer :

- Verify the type of ATM machine, if it has a touch screen, both keypad

buttons only, or both.

- Verify that on properly inserting a valid card different banking options appear on the screen.
- Check that no option to continue and enter credentials is displayed to the user when the card is inserted incorrectly.
- Verify that the touch of the ATM screen is smooth and operational.
- Verify that the user is presented with the option to choose a language for further operations.
- Check that the user is asked to enter a pin number before displaying any card/bank account detail.
- Verify that there is a limited number of attempts up to which the user is allowed to enter the pin code.
- Verify that if the total number of incorrect pin attempts gets surpassed then the user is not allowed to continue further. And operations like temporary blocking of the card, etc get initiated.
- Check that the pin is displayed in masked form when entered.
- Verify that the user is presented with different account type options like- saving, current, etc.
- Verify that the user is allowed to get account details like available balance.
- Check that the correct amount of money gets withdrawn as entered by the user for cash withdrawal.
- Verify that the user is only allowed to enter the amount in multiple denominations as per the specifications.
- Verify that the user is prompted to enter the amount again in case the amount entered is less than the minimum amount configured.
- Check that the user cannot withdraw more than the total available balance and a proper message should be displayed.
- Verify that the user is provided the option to get the transaction details in printed form.
- Verify that the user's session timeout is maintained.
- Check that the user is not allowed to exceed one transaction limit amount.
Verify that the user is not allowed to exceed the one-day transaction limit amount.
- Verify that the user is allowed to do only one transaction per pin request.

38.write scenario of Microwave Oven

Answer :

- Verify that the dimensions of the oven are as per the specification provided.
- Verify that the oven's material is optimal for its use as an oven and as per the specification.
- Verify that the oven heats the food at the desired temperature properly.
- Verify that the oven heats food at the desired temperature within a specified time duration.
- Verify the ovens functioning with the maximum attainable temperature.
- Verify the ovens functioning with minimum attainable temperature.
- Verify that the oven's plate rotation speed is optimal and not too high to spill the food kept over it.
- Verify that the oven's door gets closed properly.
- Verify that the oven's door opens smoothly.
- Verify the battery requirement of the microwave oven and check that it functions smoothly at that power.
- Verify that the text written over the oven's body is clearly readable.
- Verify that the digital display is clearly visible and functions correctly.
- Verify that the temperature regulator is smooth to operate.
- Verify that the temperature regulator works correctly.
- Check the maximum capacity of the oven and test its functioning with that volume of food.
- Check the oven's functionality with different kinds of food – solid, and liquid.
- Check the oven's functionality with different food at different temperatures.
- Verify the oven's functionality with different kinds of container material.
- Verify that the power cord of the oven is long enough.
- Verify that the usage instructions or user manuals have clear instructions.



39. Write a scenario of coffee vending Machine

Answer :

- UI scenario – Verify that the dimension of the coffee machine is as per the specification.
- Verify that the outer body, as well as inner part's material, is as per the

specification.

- Verify that the machine's body color as well brand is correctly visible and as per specification.
- Verify the input mechanism for coffee ingredients-milk, water, coffee beans/powder, etc.

Verify that the quantity of hot water, milk, coffee powder per serving is correct.

- Verify the power/voltage requirements of the machine.
- Verify the effect of suddenly switching off the machine or cutting the power. The machine should stop in that situation and in power resumption, the remaining coffee should not come out of the nozzle.
- Verify that coffee should not leak when not in operation.
- Verify the amount of coffee served in single-serving is as per specification.
- Verify that the digital display displays correct information.
- Check if the machine can be switched on and off using the power buttons.
- Check for the indicator lights when the machine is switched on-off.
- Verify that the functioning of all the buttons work properly when pressed.
- Verify that each button has an image/text with it, indicating the task it performs.
- Verify that a complete quantity of coffee should get poured in a single operation, no residual coffee should be present in the nozzle.
- Verify the mechanism to clean the system works correctly- foamer.
- Verify that the coffee served has the same and correct temperature each time it is served by the machine.
- Verify that the system should display an error when it runs out of ingredients.

40. Write a scenario of a chair.

Answer :

- Verify that the chair is stable enough to take an average human load.
- Check the material used in making the chair-wood, plastic etc.
- Check if the chair's legs are level to the floor.
- Check the usability of the chair as an office chair, normal household chair.
- Check if there is back support in the chair.
- Check if there is support for hands in the chair.
Verify the paint's type and color.
Verify if the chair's material is brittle or not.
- Check if the cushion is provided with a chair or not.
- Check the condition when washed with water or the effect of water on the chair.
- Verify that the dimension of the chair is as per the specifications.
- Verify that the weight of the chair is as per the specifications.
- Check the height of the chair's seat from the floor.



40. What is the procedure for GUI testing ?

Answer :

- **Gathering Requirements** : understand the requirements for the GUI, including functionality , visual elements and user interactions.
- **Plan Tests** : develop test cases that cover all aspects of the GUI , including positive and negative scenarios.

- **Execute Tests** : Run the test and compare the actual results with the expected result.
 - **Report defect** : Document any issue or defects found during testing.
 - **Review and validate** : review the defects and ensure that they are fixed correctly.
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41. When to use Usability Testing ?

Answer : Aesthetics and design are important. How well a product looks usually determines how well it works.

There are many software applications / websites , which miserably fail, once launched due to following reasons -

- Where do I click next ?
- Which pages need to be navigated ?
- Which Icon or jargon representative what ?
- Error message are not consistent or effectively displayed
- Session time not sufficient

Usability testing identifies usability errors in the system early in development Cycle and can save a product from failure.

42. Write a scenario of wrist Watch

Answer :

- Verify the type of watch – analog or digital.
- In the case of an analog watch, check the correctness time displayed by the second, minute, and hour hand of the watch.
- In the case of a digital watch, check if the digital display for hours, minutes, and seconds is correctly displayed.
- Verify the material of the watch and its strap.
- Check if the shape of the dial is as per specification.
- Verify the dimension of the watch is as per the specification.
- Verify the weight of the watch.
- Check if the watch is waterproof or not.
- Verify that the numbers in the dial are clearly visible or not.
- Check if the watch is having a date and day display or not.
- Verify the color of the text displayed in the watch – time, day, date, and other information.
- Verify that the clock's time can be corrected using the key in case of an

analog clock and buttons in case of a digital clock.

- Check if the second hand of the watch makes a ticking sound or not.
 - Verify the brand of the watch and check if it's visible in the dial.
 - Check if the clock is having stopwatch, timers, and alarm functionality or not.
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43. Write a scenario of Lift.

Answer :

- Verify the dimensions of the lift.
 - Verify the type of door of the lift is as per the specification.
 - Verify the type of metal used in the lift interior and exterior.
 - Verify the capacity of the lift in terms of the total weight.
 - Verify the buttons in the lift to close and open the door and numbers as per the number of floors.
 - Verify that the lift moves to the particular floor as the button of the floor is clicked.
 - Verify that the lift stops when the up/down buttons on a particular floor are pressed.
 - Verify if there is an emergency button to contact officials in case of any mishap.
 - Verify the performance of the floor – the time taken to go to a floor.
 - Verify that in case of power failure, the lift doesn't free-fall and gets halted on the particular floor.
 - Verify lifts working in case the button to open the door is pressed before reaching the destination floor.
 - Verify that in case of power failure, the lift doesn't free-fall and gets halted on the particular floor.
 - Verify lifts working in case the button to open the door is pressed before reaching the destination floor.
 - Verify that in case the door is about to close and an object is placed between the doors if the doors sense the object and again open or not.
 - Verify the time duration for which the door remains open by default.
 - Verify if the lift interior is having proper air ventilation.
 - Verify lighting in the lift.
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44. Difference between priority and severity

Answer :

<u>Severity</u>	<u>Priority</u>
Make the order in which defects should be fixed as per requirement of business.	Indicates the impact of defects on the application's functionality.
Severity is absolute and Customer-Focused.	Priority is Relative and Business-Focused.
Types: low,medium,high,critical.	Types: major,minor,moderate,cosmetic,critical

45.what is priority ?

Answer : Indicates the impact of defects on the application's functionality and Priority is Relative and Business-Focused.

- Types of Priority: - Major - Minor - Moderate - Cosmetic - Critical

46.what is severity ?

Answer : Make the order in which defects should be fixed as per requirement of business and Severity is absolute and Customer-Focused

- Types of severity: - Low - Medium - High - Critical

47. Bug categories are

Answer : -

Functional bugs
Performance bugs -
Usability bugs -
Security bugs -
UI bugs -
Regression bugs

48. Write a Scenario of whatsapp Group (generate group).

Answer : - Check the purpose of making a group.

- Check how many members are there as per requirement.
 - Check how many members can send a message or only the admin can send a message.
 - How many admins are there in the group?
 - Check the photo photoworks properly.
 - Check the emoji option works properly.
 - What is the duration of disappearing messages?
 - Check group permissions button works properly.
 - Check we can add members after creating a group.
 - Check call and video call functionality works properly.
 - Check the edit name of group functionality works properly.
 - Check remove members functionality works properly or not.
 - Check report and exit group functionality works properly.
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49. Write a scenario of only Whatsapp chat messages.

- Verify that opening any chat works or not.
- Verify that closing any chat works or not.
- Verify that sending messages works or not.
- Verify that images & videos are sent or not.
- Verify that images & videos are sent in different types of work or not.
- Verify that sending emojis works or not.
- Verify that sending stickers works or not.
- Verify that sending gifts works or not.
- Verify that creating avatar works or not.
- Verify that sending avatar works or not.
- Verify if the change is working or not.
- Verify that text editing works or not.
- Verify if the shared Gboard works or not.
- Verify that floating works or not.
- Verify that the resize works or not.
- Verify if the next language works or not.
- Verify that undoing works or not.
- Verify that the send location works or not.
- Verify that the audio sent works or not.
- Verify that AI images work or not.
- Verify that poll is works or not
- Verify that the contact sent works or not.
- Verify that the audio message is recorded and sent or not.
- Verify that send message confirmation is shown or not.
- Verify that disappearing messages work or not.
- Verify that after a long press on the space button, change language works or not.
- Verify that the message forward works or not.
- Verify that send message sent date & time shows or not.
- Verify if the pin message works or not.
- Verify that send message delete or delete for everyone works or not.
- Verify that create and add to this list works or not.
- Verify that Clear chat works or not.
- Verify that block contact works or not.

- Verify that the react message works or not.
 - Verify that send message can give a star i.e. works or not.
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